GEMS ozone evaluation using consecutive summertime ozonesonde measurements during ACCLIP

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Introduction

The Geostationary Environment Monitoring Spectrometer (GEMS), launched in February 2020, is the first geostationary instrument dedicated to atmospheric composition over Asia (*Kim et al., 2020*). The GEMS ozone profile (O3P) algorithm is continuously evolving (v2 → v3, *Bak et al., 2026*). Each release must be quantitatively assessed against independent vertical measurements. Ozonesonde measurements remain the primary reference standard for satellite validation, providing ~100 m vertical resolution with well-characterized uncertainties from the surface to ~35 km (*Smit et al., 2024*; *Thompson et al., 2017*). Here we use 62 daily ozonesonde profiles from the pre-ACCLIP and ACCLIP campaigns to quantify how much GEMS version 3 improves upon version 2 in reproducing daily ozone variability during the Asian summer monsoon.

Table 1. Summary of the datasets used in this study.

Data	Platform / Site	Product (Version)	Spatial res.	Temporal res.	Vertical res.
ECC Ozonesonde	Anmyeon (AMY) (36.54°N, 126.33°E)	O3 (ppbv, mPa) and atmospheric vars (P, T, RH, wind, etc.)	Point (*in situ*)	Daily, 8/5 – 8/31 in 2021 7/27 – 9/2 in 2022	Surface – 35km, ~100m
GEMS	GEO-KOMPSAT-2B (Geostationary)	O3P L2 (v2, v3)	~ 14 x 31 km²	Hourly (00 – 07 UTC)	24 layers,
TROPOMI	Sentinel-5P (LEO, ~13:30 LT)	O3 profile L2	28 x 28 km²	Daily	33 layers
ERA5	ECMWF Reanalysis	Pressure level	1.5° x 1.5°	6 hourly	37 layers
IMERG	GPM (Multi-satellite)	Final Run V07B	0.1° x 0.1°	30 min	Surface

Result 1

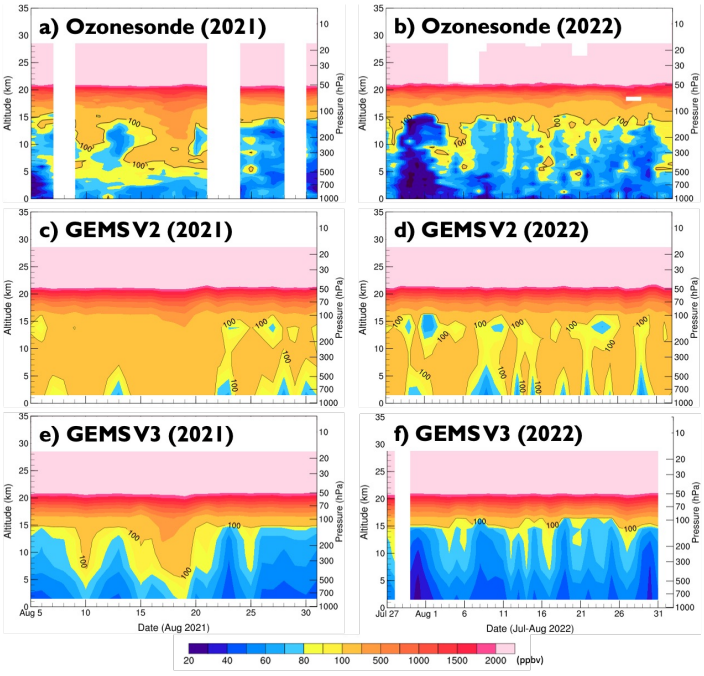


Figure 2. Time–height cross sections of ozone mixing ratio (shading, unit: ppbv) from ozonesonde measurements (a, b), GEMS v2 (c, d), and GEMS v3 (e, f). The left (a, c, e) and right (b, d, f) panels correspond to data from 2021 and 2022, respectively.

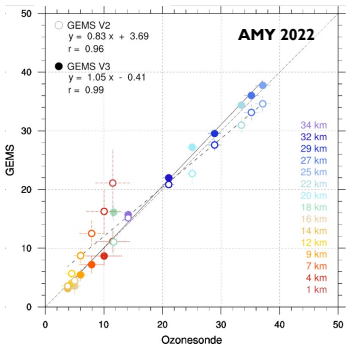


Figure 3. Scatter plots with regression lines of column-integrated ozone (unit: DU) over the GEMS vertical layers for (a) 2021 and (b) 2022. The open and closed circles represent GEMS v2 and GEMS v3, respectively. Points are colored by layer altitude, from reddish at lower altitudes to bluish at higher altitudes. Whiskers denote ±1σ standard deviation.

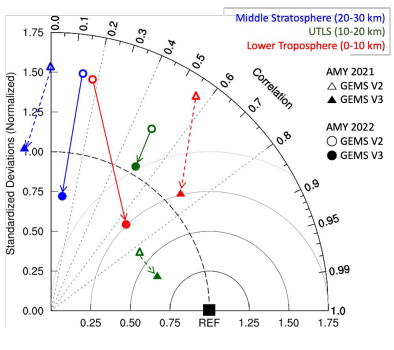


Figure 4. Taylor diagram of the mean column ozone in the LT (red), UTLS (green), and MS (blue) for 2021 (triangles) and 2022 (circles). Open and closed markers represent GEMS v2 and GEMS v3, respectively. The black square indicates the reference normalized standard deviation (=1.00). Arrows indicate the change from GEMS v2 to GEMS v3. LT, lower troposphere; UTLS, upper troposphere lower stratosphere; MS, middle stratosphere.

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Ozone observation

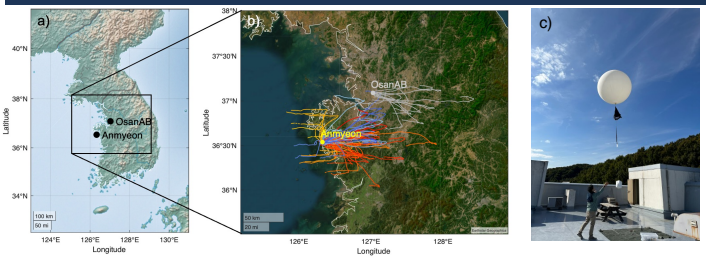


Figure 1. (a) Locations of Anmyeon and Osan Air Base (OsanAB). (b) Ascending trajectories of ozonesondes. Bluish and reddish lines represent the Anmyeon measurements in 2021 and 2022, respectively. Gray lines show OsanAB measurements in 2022. (c) Example of launching ozonesonde with meteorological balloon.

Table 2. Anmyeon soundings are used in this study; Yongin and Osan AB are listed for campaign context (*Kang et al., 2024*).

Sation	Latitude (°)	Longitude (°)	Elevation (m)	Record (month/year)	Number of profiles (#)
Anmyeon	36.54	126.33	8	Aug / 2021	24
				Jul-Sep / 2022	38
Yongin	37.33	127.26	135	Oct-Nov / 2021	10
				May-Jun / 2022	17
				Aug / 2023	12
				Sep-Oct / 2024	12
				Aug-Sep / 2025	17
OsanAB	37.09	127.03	10	Aug / 2022	11

Result 2

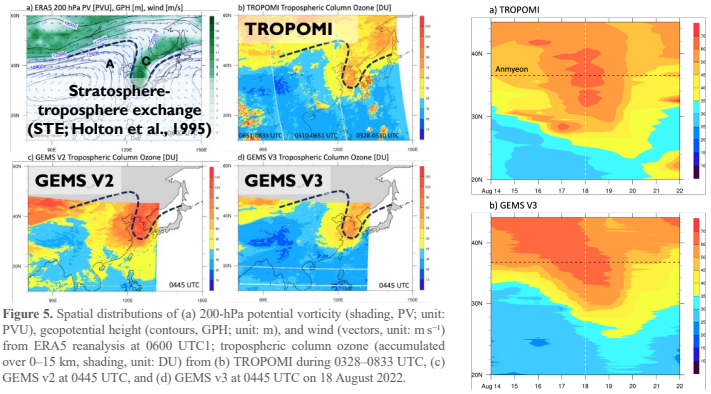


Figure 5. Spatial distributions of (a) 200-hPa potential vorticity (shading, PV; unit: PVU), geopotential height (contours, GPH; unit: m), and wind (vectors, unit: m s⁻¹) from ERA5 reanalysis at 0600 UTC1; tropospheric column ozone (accumulated over 0–15 km, shading, unit: DU) from (b) TROPOMI during 0328–0833 UTC, (c) GEMS v2 at 0445 UTC, and (d) GEMS v3 at 0445 UTC on 18 August 2022.

Figure 6. Time–latitude cross sections of tropospheric column ozone (0–15 km, unit: DU) averaged over longitudes 120–130°E for the period from 14–22 August 2021: (a) TROPOMI and (b) GEMS v3. The black dashed lines indicate the latitude of the Anmyeon site (36.54°N), and the white dashed lines represent 14 August 2021, when enhanced ozone intrusion was evident over South Korea.

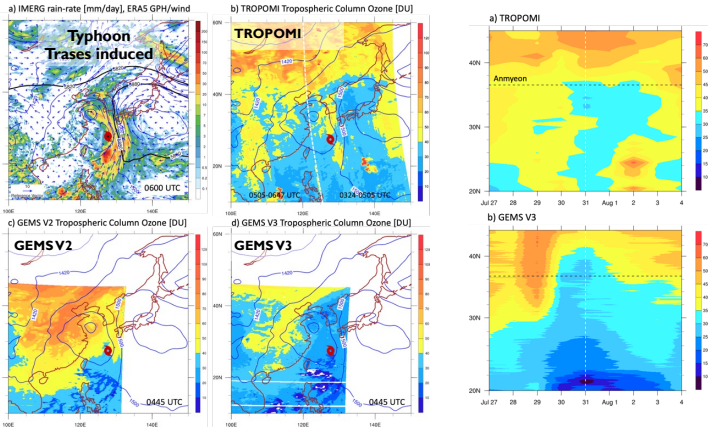


Figure 7. Spatial distributions of (a) integrated multi-satellite retrievals for global precipitation measurement (IMERG) daily rain-rate (shading, unit: mm day⁻¹), 850 hPa GPH (blue contours, unit: m), 500 hPa geopotential height (black contours, unit: m), and wind (vectors, unit: m s⁻¹); and tropospheric column ozone (accumulated over 0–15 km, shading, unit: DU) from (b) TROPOMI during 0324–0647 UTC, (c) GEMS v2 at 0445 UTC, and (d) GEMS v3 at 0445 UTC on 31 July 2022.The typhoon marker indicates the position of Typhoon Trases at 0600 UTC on 31 July 2022 (latitude: 27.6°N, longitude: 127.6°E).

Figure 8. Time–latitude cross sections of tropospheric column ozone (0–15 km, unit: DU) averaged over longitudes 120–130°E for the period from 27 July to 4 August 2022: (a) TROPOMI and (b) GEMS v3. The black dashed lines indicate the latitude of the Anmyeon site (36.54°N), and the white dashed lines represent 31 July 2022, when typhoon Trases approached South Korea.

Conclusion

- GEMS v3 reproduces the observed day-to-day ozone variability both in profile shape and in regression statistics, with the largest improvement in the troposphere.
- GEMS v3 resolves the evolving spatial distribution of ozone during both the STE and typhoon events, consistent with the ozonesonde profiles.
- Continuous ozonesonde measurements enable satellite evaluation on synoptic timescales and across a broad range of weather events.